

Advanced SQL Date Functions

Q1: Add 6 months to each hire date.

```
SELECT emp_id, name, hire_date  
DATEADD(month, 6, hire_date) AS hire_plus_6_months  
FROM employees;
```

Outcome

emp_id	name	hire_date	hire plus 6 months
1	Alice	2020-1-15	2020-07-15
2	Bob	2021-6-10	2021-12-10
3	Charlie	2023-3-22	2023-09-22

Q2: Age in days from dob to today.

```
SELECT student_id, name  
DATEDIFF(day, dob, CURRENT_DATE) AS age_in_days  
FROM students;
```

Outcome

student_id	name	age_in_days
101	Maya	7,578
102	Ethan	7,800
103	Sienna	7,000

Q3: Days remaining until each event.

```
SELECT event_id, event_name  
DATEDIFF(day, CURRENT_DATE, event_date) AS days_remaining  
FROM events;
```

event_id	event_name	days_remaining
1	Seminar	negative (past event)
2	Workshop	future positive value
3	Hackathon	future positive value

Q4: Days between issue_date and due_date
 SELECT invoice_id, issue_date, due_date,
 DATEDIFF(day, issue_date, due_date) AS days_between
 FROM Invoices;

Outcome:

invoice_id	issue_date	due_date	days_between
501	2023-03-10	2023-03-25	15
502	2023-04-01	2023-04-15	14
503	2023-04-10	2023-04-20	10

Q5: Format start_date as Month YYYY-TO-CHAR()
 SELECT course_id, name,
 TO_CHAR(start_date, 'Month YYYY') AS formatted_date
 FROM courses;

Outcome

course_id	name	formatted_date
201	SQL Basics	May 2023
202	Python	June 2023

Q6: Create full date from parts.
 SELECT member_id,
 DATE_FROM_Parts(start_year, start_month, start_day) AS full_start_date
 FROM membership;

Outcome

member_id	full_start_date
1	2023-05-10
2	2022-11-25

Q7: Extend renewal_date by 1 year.

```
SELECT sub_id, plan  
DATEADD(year, 1, renewal_date) AS extended_renewal_date  
FROM subscriptions;
```

Outcome.

Sub_id	Plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Q8: Current date & days since order

```
SELECT order_id, order_date,  
CURRENT_DATE AS Today_date,  
DATEDIFF(day, order_date, CURRENT_DATE) AS days_since_order.  
FROM orders;
```

Outcome.

order_id	order_date	today_date	days_since_order
1001	2025-04-13	2026-05-10	390
1002	2025-04-10	2026-05-10	395

Q9: Extract year from training_date

```
SELECT training_id, topic,  
EXTRACT(year, FROM training_date) AS training_year  
FROM training;
```

Outcome.

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

Q10: Extract hour and minute from published_on (4)

```
SELECT post_id, title,
EXTRACT(hour FROM published_on) AS hour_published,
EXTRACT(minute FROM published_on) AS minute_published,
FROM Blog_Posts;
```

Outcome.

post_id	title	hour_published	minute_published
1	SQL Tips	10	15
2	Data Cleaning	10	45

Q11: Days until license expiry

```
SELECT driver_id, license_expiry,
DATEDIFF(day, CURRENT_DATE, license_expiry) AS days_left
FROM Drivers;
```

Outcome

driver_id	license_expiry	days_left
301	2025-08-10	negative
302	2023-12-31	negative

Q12: Seconds since message was sent.

```
SELECT message_id, sent_timestamp,
CURRENT_TIMESTAMP AS current_timestamp,
DATEDIFF(second, sent_timestamp, CURRENT_TIMESTAMP) AS
seconds_since_sent.
FROM messages;
```

Outcome

message_id	sent_timestamp	current_timestamp	seconds_since_sent
1	2025-04-19 09:32:45	(now)	computed value
2	2025-04-18 23:59:59	(now)	computed value

Q13: Add 15 days to return_date

```
SELECT return_id, return_date  
DATEADD(day, 15, return_date) AS restock_date  
FROM returns;
```

Outcome

return_id	return_date	return_date restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Q14: Convert string to date.

```
SELECT assign_id,  
TO_DATE(assigned_on, 'YYYY-MM-DD') AS Assign_on_date  
FROM Assignments;
```

Outcome.

assign_id	assigned-on-date
1	2025-03-01
2	2025-03-05

Q15: Format timestamp as readable string

```
SELECT meeting_id,  
TO_CHAR(scheduled_time, 'FMMonth DD, YYYY "at" HH:MI:AM') AS  
Formatted_meeting_time  
FROM Meetings;
```

Outcome

meeting_id	formatted-meeting-time
1	April 19, 2025 at 2:00 PM
2	April 19, 2025 at 9:30 AM